Year 10 Mathematics

Algebra 2

7.1 Indices

Indices

STARTER · 5 MIN

Date: _____ **Title:** Indices

LEARNING OBJECTIVES

- Use the laws of indices with algebraic terms
- Simplify expressions with brackets and powers
- Use negative and fractional indices with letters

STARTER *Answer in your book. No calculators.*

1. Work out $2^3 \times 2^2$

2. Work out 5^0

3. Work out 3^{-2}

4. Work out $\sqrt{81}$

Starter — Answers

STARTER

Tick or correct in green pen

1. $2^5 = \mathbf{32}$

2. $5^0 = \mathbf{1}$

3. $\frac{1}{3^2} = \frac{\mathbf{1}}{\mathbf{9}}$

4. $\sqrt{81} = \mathbf{9}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the laws of indices with algebraic terms	☐
SILVER	I can simplify expressions with brackets and powers	☐
GOLD	I can use negative and fractional indices with letters	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Laws of Indices

INTRODUCTION · 3 MIN

The same rules work for letters as for numbers.

Multiply

Add the indices.

$x^4 \times x^3 = x^7$

Divide

Subtract the indices.

$y^8 \div y^5 = y^3$

Power of a power

Multiply the indices.

$(a^3)^4 = a^{12}$

Negative and fractional

One over, and roots.

$x^{-2} = \frac{1}{x^2}, \quad x^{\frac{1}{2}} = \sqrt{x}$

Examples 1 and 2 — Multiplying and Dividing

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Example 1

Simplify

$3x^4 \times 5x^2$

Example 2

Simplify

$18y^7 \div 6y^3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$3 \times 5 \times x^{4+2}$

$= 15x^{6}$

Tip: multiply the numbers, add the indices.

Example 2

$18 \div 6 \times y^{7-3}$

$= 3y^{4}$

Tip: divide the numbers, subtract the indices.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify each expression*

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

1 $x^5 \times x^3$

2 $a^9 \div a^4$

3 $2p^3 \times 4p^2$

4 $20m^6 \div 5m^2$

5 $3x^2y \times 2xy^3$

★ $x^4 \times x^0$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $x^{5+3} = \mathbf{x^{8}}$

2 $a^{9-4} = \mathbf{a^{5}}$

3 $8p^{3+2} = \mathbf{8p^{5}}$

4 $4m^{6-2} = \mathbf{4m^{4}}$

5 $6x^{2+1}y^{1+3} = \mathbf{6x^{3}y^{4}}$

★ $x^{4+0} = \mathbf{x^{4}}$

Examples 3 and 4 — Brackets and Powers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $(a^m)^n = a^{mn}$

Example 3

Simplify

$(2x^3)^4$

Example 4

Simplify

$$\frac{(3a^2)^3}{9a^4}$$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$2^4 \times x^{3 \times 4}$

$= 16x^{12}$

Tip: raise the number to the power too.

Example 4

$$\frac{27a^6}{9a^4}$$

$= 3a^2$

Tip: expand the bracket first, then divide.

Task 2

PRACTICE · 9 MIN

SILVER *Simplify each expression*

Hint $(a^m)^n = a^{mn}$

1 $(x^4)^3$

2 $(3y^2)^2$

3 $(2a^3b)^3$

4 $\dfrac{(4x^3)^2}{8x^2}$

5 $\dfrac{6p^5 \times 2p^3}{4p^6}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $x^{4\times3} = \mathbf{x^{12}}$

2 $3^2y^{2\times2} = \mathbf{9y^4}$

3 $2^3a^9b^3 = \mathbf{8a^9b^3}$

4 $\dfrac{16x^6}{8x^2} = \mathbf{2x^4}$

5 $\dfrac{12p^8}{4p^6} = \mathbf{3p^2}$

Examples 5 and 6 — Negative and Fractional Indices

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\quad a^{\frac{1}{2}} = \sqrt{a}$

Example 5

(2 marks)

Simplify, giving your answer with a positive index:

$x^3 \div x^7$

Example 6

(2 marks)

Simplify

$(16x^8)^{\frac{1}{2}}$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$x^{3-7} = x^{-4}$$

$$= \frac{1}{x^4}$$

Tip: a negative index means one over.

Example 6

$$\sqrt{16} \times x^{8 \times \frac{1}{2}}$$

$$= 4x^4$$

Tip: power one half: square root the number, halve the index.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\quad a^{\frac{1}{2}} = \sqrt{a}$

1. Simplify $a^2 \div a^5$ (1 mark)

2. Simplify $(25y^6)^{\frac{1}{2}}$ (2 marks)

3. Simplify $(8x^9)^{\frac{1}{3}}$ (2 marks)

4. Simplify $3x^{-2} \times 4x^5$ (2 marks)

5. Find n: $x^n \times x^4 = x^{-2}$ (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $a^{-3} = \dfrac{1}{\mathbf{a^3}}$

2 $\sqrt{25}\,y^3 = \mathbf{5y^3}$

3 $\sqrt[3]{8}\,x^3 = \mathbf{2x^3}$

4 $12x^{-2+5} = \mathbf{12x^3}$

5 $n + 4 = -2, \quad n = \mathbf{-6}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $4x^3 \times 2x^5$

SILVER Simplify $(3y^4)^2$

GOLD Simplify $(9x^4)^{\frac{1}{2}}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $8x^{3+5} = \mathbf{8x^{8}}$

SILVER $3^{2}y^{8} = \mathbf{9y^{8}}$

GOLD $\sqrt{9}\,x^{2} = \mathbf{3x^{2}}$

Exam Practice — Multiplying and Dividing

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Question 1

(2 marks)

Simplify

(a) (1)

$x^5 \times x^3$

.....

(b) (1)

$y^9 \div y^4$

.....

Question 2

(4 marks)

Simplify

(a) (2)

$3a^4 \times 5a^2$

.....

(b) (2)

$$\frac{20b^7}{4b^3}$$

.....

Exam Practice — Brackets and Powers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $(a^m)^n = a^{mn}$

Question 3

(3 marks)

Simplify

(a) (1)

$(p^3)^5$

.....

(b) (2)

$(2x^4)^3$

.....

Question 4

(3 marks)

Simplify fully

$$\frac{(2x^3)^4}{4x^5}$$

.....

Exam Practice — Negative and Fractional Indices

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^{-n} = \frac{1}{a^n}$ $\qquad a^{\frac{1}{2}} = \sqrt{a}$

Question 5

(3 marks)

(a) Simplify, with a positive index (1)

$$\frac{x^2}{x^6}$$

.....

(b) Simplify (2)

$$(16x^8)^{\frac{1}{2}}$$

.....

Question 6

(2 marks)

Simplify fully

$$(27y^6)^{\frac{2}{3}}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) x^8 B1

(b) y^5 B1

Tip: × add the powers, ÷ subtract them.

Question 2

(a) $15a^6$ B2

B1 for 15 or a^6

(b) $5b^4$ B2

B1 for 5 or b^4

Tip: deal with numbers and letters separately.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) p^{15} B1

(b) $8x^{12}$ B2

B1 for 8 or x^{12}

Tip: the power applies to everything inside.

Question 4

$(2x^3)^4 = 16x^{12}$ M1

$16x^{12} \div 4x^5$ M1

$= 4x^7$ A1

Tip: $2^4 = 16$, not 8.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $\mathbf{x^{-4} = \frac{1}{x^4}}$ — **B1**

(b) $\sqrt{16} = 4$, $x^8 \rightarrow x^4$ — **M1**

$\mathbf{4x^4}$ — **A1**

Tip: a power of a half halves every power inside.

Question 6

$\sqrt[3]{27} = 3$, $3^2 = 9$ — **M1**

$\mathbf{9y^4}$ — **A1**

Tip: two-thirds of the power of y: 6 → 4.